Original Article

# The Effect of Water Vapor Content on the Reynolds Number of an Air-Water Vapor Mixture Flowing through a Horizontal Air Duct

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## ABSTRACT

The following article presents a study on the effect of the water vapor content in air, considered as a two-phase mixture, on the Reynolds number during the flow of an air–water vapor mixture through a horizontal air duct. More specifically, the article provides an analytical description of the mathematical model that relates the change in the Reynolds number of the air–water vapor mixture flow to the relative humidity of the water vapor. The study focused on the possibility of controlling the Reynolds number of airflow solely by adjusting the relative humidity of the water vapor in the air. The mathematical model was applied for air velocities ranging from 0 to 128 m/s, relative humidity levels from 0% to 100%, and air temperatures from -40 °C to 150 °C. The figures and study's results indicate that increasing the relative humidity level of the water vapor leads to a decrease in the Reynolds number of the air–water vapor mixture as it flows through an air duct. Additionally, it is possible to control the Reynolds number of the flowing air–water vapor mixture solely through the relative humidity of the water vapor in the mixture. This capability is particularly useful in a wide range of industrial processes.

## Introduction

The Reynolds number is a fundamental dimensionless parameter in fluid mechanics. It characterizes the type of fluid flow-specifically, whether the flow is laminar or turbulent. In laminar flow, the fluid streamlines move in parallel to the axis of the internal surface of the horizontal air duct. In contrast, during turbulent flow, the streamlines exhibit irregular and chaotic motion. Regarding the Reynolds number range and corresponding flow types: When Re < 2300, the flow is considered laminar. When Re > 2900, the flow becomes turbulent. For values between 2300 < Re < 2900, the flow is in a transitional state, where streamlines shift from orderly (laminar) to chaotic (turbulent) motion [1,2]. The Reynolds number for a fluid flowing through an air duct is calculated using Equation 1a.

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$$Re = \frac{\rho u D_h}{\mu} \quad (1a)$$

The above equation allows for the study of Reynolds number variation in homogeneous fluids based on parameters such as flow velocity, fluid density, fluid viscosity, and the diameter of the air duct. However, it does not account for mixtures—*i.e.* fluids composed of different phases or varying compositions, such as air containing water vapor. In the case of a mixture consisting of two or more components, calculating the Reynolds number becomes more complex. Therefore, Equation 1a can be modified to Equation 1b, as follows:

$$Re = \frac{\rho_m u D_h}{\mu_m} = \frac{\rho_m(\Phi) u(\Phi)}{\mu_m(\Phi)} D_h \quad (1b)$$

Using Equation 1b, the Reynolds number of a mixture and its components can be expressed as a function of the densities and viscosities of those components. This equation can also be applied to atmospheric air and its individual gaseous constituents. Regarding the flow velocity of the air–water vapor mixture, for the sake of simplification, it is assumed that the velocity of the water vapor component is equal to that of the air. In general, it is assumed that during the flow of a mixture, all individual components move at the same velocity.

The mathematical model examined in this article allows us to investigate how the Reynolds number and the type of airflow in a piping system can be controlled in relation to the relative humidity of the water vapor amount contained in the air. This capability is particularly useful in industrial applications and other processes that specifically require laminar, turbulent, or transitional flow conditions. In such cases, the Reynolds number of the airflow should be pre-adjustable within certain limits or set to specific values, depending on the needs of each application or process. The resulting mathematical model can be applied in research across a wide range of scientific and engineering fields, such as pharmaceuticals, HVAC systems, mechanical and aerospace engineering, computational fluid dynamics (CFD), fluid mechanics, chemical engineering, biomedical engineering, public health, and food processing. As an example, Table 1 presents industrial applications, procedures, and other processes that achieve optimal performance under laminar airflow conditions.

Table 2 reports industrial applications, procedures, and other processes that have optimal performance during the transitional state of the airflow.

### *Review of Literature*

Ahmed S. *et al.* conducted an experiment using a horizontal rectangular air duct to study the influence of the Reynolds number and air velocity on the evaporation rate of water vapor. They also examined the dimensionless Nusselt and Sherwood numbers of humid air for various Reynolds number values in the range of 11,000 < Re < 41,000 [45]. According to the results, an increase in the Reynolds number of the flow leads to an increase in both the Nusselt and Sherwood numbers.Ahmed A. *et al*. conducted a CFD study on a cross-flow heat exchanger with staggered elliptical tubes. The study examined how changes in Reynolds number and relative humidity jointly affect the Nusselt number during convective heat transfer. The relative humidity in the study ranged from 0% to 80% at 323 K. According to the study, an increase in the Reynolds number of the airflow leads to an increase in the Nusselt number. Additionally, as relative humidity increases, the Nusselt number also increases. It is already known that these two dimensionless numbers are correlated—an increase in relative humidity leads to an increase in the Reynolds number of the airflow [46].

**Table 1** Applications and procedures that require laminar airflow

| Sector | Application / Procedure | Ref. |
|---|---|---|
| Pharmaceutical manufacturing | Laminar flow and low Reynolds numbers are crucial for maintaining product purity and sterility, particularly in sterile filling operations and cleanrooms. These factors are essential for preventing microbial growth and ensuring consistent product quality. | [33] |
| HVAC systems in health sector | Low air velocity, Reynolds number and laminar airflow in the air ducts lead to an increase in the particle collection efficiency of HEPA filters in the HVAC systems; apart from that, enhance the ability of UVGI lamps to kill pathogens and increase the sterilization efficiency of lamps. | [34,35] |
| Biology and biomedical engineering | Laminar flow helps with microbial control. It can prevent biofilm formation by microorganisms in liquid storage tanks, highlighting the need for proper cleaning and sanitization procedures. Additionally, laminar flow facilitates the study of cell behavior in controlled microenvironments as well as the creation of concentration gradients that can mimic in vivo conditions, enabling researchers to study cellular responses to various stimuli. | [33,35] |
| Chemicals and materials production | Laminar flow enables the creation of complex structures and materials by controlling the mixing and deposition of different substances. This is particularly useful in the synthesis of micro- and nanoparticles, and other materials with specific properties. Additionally, laminar flow is used to generate and manipulate droplets, which are helpful for encapsulating materials, performing reactions, and delivering drugs. The predictable flow allows for the generation of droplets with consistent sizes and properties. In addition, although laminar flow is characterized by slow mixing, it can be utilized to enhance mass transfer by controlling the phases of the fluids. This is particularly useful in applications where diffusion is the selected mechanism for mass transfer. | [36-39] |
| Air pumping systems | Laminar flow is important for applications needing precise flow rate control and low pressure losses. The low pressure is an important factor that causes lower friction losses compared to turbulent flow. As a consequence this results in lower pumping power requirements, saving energy and reducing operational costs. When a fluid is in laminar flow, minimizes wall erosion and reduces vibration and noise. | [40] |
| Food processing | Laminar flow and precise pressure lead to maintaining sterile environments for food processing, food preservation and aseptic packaging under safe conditions. | [41] |

**Table 2** Applications and procedures that require transitional airflow

| Sector | Application / Procedure | Ref. |
|---|---|---|
| Pharmaceutical manufacturing | Dry powder inhalers can offer optimal performance for aerosol dispersion during the transitional state of the flow. The drug in the inhaler is considered as a two-phase mixture. | [42] |
| HVAC systems | Certain HVAC systems use pulsed or cyclic air flows via LTG transient flow units. These units are a type of air distribution system that utilizes pulsed airflow to enhance thermal comfort conditions of air and energy efficiency in buildings without sacrificing acoustics. Additionally, a well-designed drainage venting piping system, requires CFD transient modeling for the transitional state to prevent air hammer effects, trapped air surges, and pressure spikes. | [43] |
| Gas turbines | Startup, shutdown, load changes, and heat soak produce time-dependent aerothermal dynamics requiring precise transient modeling for reliability and fault detection. | [44] |

Another CFD study, conducted by Abid A. *et al.*, examined how different values of the Reynolds number affect airflow characteristics, thermal comfort conditions, and humidity transport dynamics within an HVAC cabinet. An experiment carried out by Cao J. *et al.* on a gas turbine investigated the influence of mist droplet and air mass ratios on the Nusselt number ratio between cooling mist droplets and cooling air in a two-phase air–water vapor mixture at 880 K. These ratios can be correlated with both the Reynolds number and the water vapor amount of the air–water vapor mixture. The study found that an increase in the mist-to-air mass ratio leads to an increase in the Nusselt number ratio. Additionally, an increase in the water vapor amount in the two-phase mixture results in a higher Reynolds number. As the Reynolds number increases, both the air mass flow rate and the mist mass flow rate increase, as does the ratio between the Nusselt number of the cooling mist droplets and that of the cooling air [47].

Two other articles investigate the effect of water vapor relative humidity on the mass transfer coefficient in a rectangular air duct [49,50]. One of the curves from these studies shows that an increase in the relative humidity of the air–water vapor mixture leads to a decrease in the Sherwood number of the airflow. Furthermore, an increase in the Reynolds number of the airflow results in an increase in the Sherwood number [45,49]. However, as the Reynolds number increases, the Sherwood number increases in an irreversible manner [45,51]. The conclusion of this research is that an increase in the relative humidity of water vapor leads to a decrease in the Reynolds number of the airflow [49].

According to Westhoff *et al.* as the Reynolds number of an air–water vapor mixture increases the Nusselt number of the flow also increases. An increase in the relative humidity of water vapor in air leads to a linear increase in the Sherwood number for various Reynolds number values. Additionally, an increase in the Reynolds number of the airflow results in a linear increase in the variation ratio between the Sherwood number and relative humidity, expressed as $dSh/d\Phi$. This indicates that a change in the relative humidity of water vapor does not necessarily affect the Reynolds number of the airflow. However, an increase in the relative humidity of water vapor is inversely proportional to the Reynolds number of the air–water vapor mixture, but only under the condition of a constant Sherwood number [52].

## Materials and Methods

### *The Density of Air – Water Vapor Mixture*

In the case of a mixture composed of n individual components, the density of the mixture is equal to the sum of the partial densities of each component or, equivalently, the sum of the products of each component's density and its mole fraction [3].

$$\rho_T = \sum_{i=1}^{x,\rho} x_1 \rho_1 \tag{2}$$

In the case of an air – water vapor mixture, Equation 2 takes the following form:

$$\rho_m = x_d \rho_d + x_v \rho_v \tag{3}$$

The mole fractions of air and water vapor are calculated by Equations 4 and 5.

$$x_d = \frac{n_d}{n_d + n_v} = \frac{\frac{m_d}{M_{rd}}}{\frac{m_d}{M_{rd}} + \frac{m_v}{M_{rv}}} \quad (4)$$

$$x_v = \frac{n_v}{n_d + n_v} = \frac{\frac{m_v}{M_{rv}}}{\frac{m_d}{M_{rd}} + \frac{m_v}{M_{rv}}} \quad (5)$$

The molecular weight of air – water vapor mixture is calculated by Equation 6:

$$M_{rm} = x_d M_{rd} + x_v M_{rv} = \frac{M_{rd} M_{rv} (m_d + m_v)}{m_d M_{rv} + m_v M_{rd}} \quad (6)$$

By substituting Equations 4-6 into Equation 3, the latter takes the following final form:

$$\rho_m = \frac{M_{rm}}{m_d + m_v} (n_d \rho_d + n_v \rho_v) \quad (7)$$

The densities of dry air $\rho_d$ and water vapor $\rho_v$ can be expressed as functions of the partial pressures of each component using the ideal gas law, assuming that the individual components of the mixture are ideal gases. Water vapor can be considered an ideal gas at pressures below 10 kPa [31]. Consequently, Equation 6 takes the following form:

$$\rho_m = \frac{M_{rm}}{m_m RT} (m_d P_d + m_v P_v) \quad (8)$$

Equation 8 can be transformed into Equation 9 according to Dalton's law for partial pressures of the individual components in the mixture (see Equation 20):

$$\rho_m = \frac{M_{rm}}{m_m RT} [m_d (P_m - P_v) + m_v P_v] \quad (9)$$

Also, Equation 8 can be transformed into the following two alternative forms [4,5]:

$$\rho_m = \frac{P_m}{RT} \left( M_{rd} \frac{P_m - P_v}{P_m} + M_{rv} \frac{P_v}{P_m} \right) \quad (10)$$

$$\rho_m = \frac{P_m M_{rd}}{RT} \left[ 1 - x_v \left( 1 - \frac{M_{rv}}{M_{rd}} \right) \right] \quad (11)$$

In the model used in this study, Equation 9 will be applied. The parameters $m_d$, $m_v$, $x_d$, $x_v$ and $M_{rm}$ are calculated based on the specific humidity $w_S$ of the air – water vapor mixture. The specific humidity is defined as the ratio of the mass of water vapor to the mass of dry air and is determined by the dry-bulb temperature $T_d$ and relative humidity of the water vapor using a psychrometric chart.

$$w_s = \frac{m_v}{m_d} = \frac{m_v}{m_m - m_v} \tag{12}$$

Regarding the relative humidity of water vapor, it can be calculated by Equation 12.

$$\Phi = \frac{P_v}{P_S} \times 100\% \tag{13}$$

$$P_S(T_\theta) = a_4 {T_\theta}^4 + a_3 {T_\theta}^3 + a_2 {T_\theta}^2 + a_1 T_\theta + a_0 \tag{14}$$

The pressure of saturated water vapor $P_S$ is calculated using Equation 14 in units of Pa $\times 10^3$ as a function of vapor temperature [5,6]. This formula provides reliable results for temperatures in the range of 10 °C ≤ $T_\theta$ ≤ 110 °C.

Equation 15 (Clausius – Clapeyron) [17], Equation 16 (August-Roche-Magnus) [14], and Equation 17 (A. L. Buck) [15] are alternative formulas for calculating the saturated water vapor pressure.

$$P_s = 6.11 \exp\left[5321\left(\frac{1}{273} - \frac{1}{T_\theta}\right)\right] \tag{15}$$

$$P_s = 6.1094 \exp\left(\frac{17.625 T_\theta}{T_\theta + 243.04}\right) \tag{16}$$

$$P_s = 6.112 \exp\left[\left(18.69 - \frac{T_\theta}{234.5}\right)\left(\frac{T_\theta}{T_\theta + 257.14}\right)\right] \tag{17}$$

The Goff – Gratch formula in Equation 18 is a more complex approach (Goff, 1957). The pressure is expressed in Pa.

$$\begin{aligned} log P_s = {} & -7.903\left(\frac{373.15}{T_\theta} - 1\right) + 5.281 \log\left(\frac{373.15}{T_\theta}\right) \\ & -1.382 \times 10^{-7}\left[10^{11.344\left(1 - \frac{T_\theta}{373.15}\right)} - 1\right] \\ & +8.133 \times 10^{-3}\left[10^{-3.49149\left(\frac{373.15}{T_\theta} - 1\right)} - 1\right] + 1031.25 \end{aligned} \tag{18}$$

The pressure of the air – water vapor mixture flow is the supply pressure generated by the energy imparted by the air pump as the mixture flows through the air duct. Additionally, the sum of the partial pressures of each component in the mixture equals the pump's supply pressure.

$$P_m = P_Q = P_v + P_d \quad (19)$$

The supply pressure is calculated as a function of the mixture's flow velocity. Additionally, atmospheric pressure and density variations due to temperature must be taken into account.

$$P_Q = P_{atm} + \frac{1}{2}\rho_m {u_m}^2 \quad (20)$$

Equation 21 is derived by Equation 13 with respect to $P_v$ by substituting into Equation 9 [5]. The density of the air – water vapor mixture can be expressed as a function of the relative humidity of the water vapor using the following equation.

$$\rho_m(\Phi) = \frac{P_m}{RT}\left[M_{rd} + \Phi\frac{P_s}{P_m}(M_{rv} - M_{rd})\right] \quad (21)$$

The validity of Equation 21 can be tested by setting the parameters $M_{rm}$ = 0 and Φ = 0. Under these conditions, Equation 21 reduces to the ideal gas equation of state. This implies that there is no water vapor or moisture in the air duct–only dry air, which behaves as an ideal gas.

### *The Viscosity of Air – Water Vapor Mixture*

Similar to the density of the air – water vapor mixture, the viscosity of a mixture composed of n individual components is equal to the sum of partial viscosities of each component or, alternatively, the sum of the products of each component's viscosity and its mole fraction [3].

$$\mu_T = \sum_{i=1}^{v,\mu} x_1\mu_1 \quad (22)$$

In the case of the air – water vapor mixture the following equation applies:

$$\mu_m = x_d\mu_d + x_v\mu_v \quad (23)$$

Regarding the calculation of the mixture's viscosity based on the partial viscosities of each component, various mathematical formulas have been proposed, such as the Golubev equation [8], which incorporates the partial viscosities, molecular weights, and mole fractions of each component.

$$\mu_m = \frac{1}{x_v + x_d}\left(\frac{x_d\mu_d}{x_d + c_w} + \frac{x_v\mu_v}{x_v + c_w}\right) \quad (24)$$

Where, $c_w$ is Wilke's coefficient [25], calculated by Equation 25, as follows:

$$c_w = \frac{\left[1 + \left(\frac{M_{rv}}{M_{rd}}\right)^{\frac{1}{4}}\sqrt{\frac{\mu_d}{\mu_v}}\right]^2}{2\sqrt{2\left(1 + \frac{M_{rd}}{M_{rv}}\right)}} \quad (25)$$

Equations 24,25 are reliable for various air mixtures whose components have similar molecular weight. However, for mixtures with significant differences in molecular weights– such as the air – water vapor mixture, where $M_{rd} = 28.96$ gr/mol and $M_{rv} = 18$ gr/mol, respectively, representing a 37.8% deviation. Schmick's equation may provide more accurate calculations [7].

Krischer and Kast suggest an equation similar to Equation 26, in which the mole fraction is replaced by the pressure. Therefore, the viscosity of the air-water vapor mixture is calculated based on the partial pressures of each component of the mixture [4,10].

*Determination of Air Viscosity*

Sutherland's empirical formula [27] is a fundamental equation for calculating the viscosity of air, derived from the kinetic theory of gases [18]. This formula provides reliable results for temperatures ranging from 0 to 555 K and pressures up to 3.4 MPa.

$$\mu_m = \frac{x_d \mu_d \sqrt{M_{rd}} + x_v \mu_v \sqrt{M_{rv}}}{x_d \sqrt{M_{rd}} + x_v \sqrt{M_{rv}}} \tag{26}$$

$$\mu_m = \frac{\mu_d (P_m - P_v)\sqrt{M_{rd}} + \mu_v P_v \sqrt{M_{rv}}}{(P_m - P_v)\sqrt{M_{rd}} + P_v \sqrt{M_{rv}}} \tag{27}$$

By replacing of Equation 13 with Equation 27, we obtain:

$$\mu_m(\Phi) = \frac{\mu_d (P_m - \Phi P_S)\sqrt{M_{rd}} + \mu_v \Phi P_S \sqrt{M_{rv}}}{(P_m - \Phi P_S)\sqrt{M_{rd}} + \Phi P_S \sqrt{M_{rv}}} \tag{28}$$

$$\mu_d(T) = \frac{1.458 \times 10^{-6}\, T^{\frac{3}{2}}}{110.4 + T} \tag{29}$$

Equation 29 can be modified by adding an additional coefficient, resulting in accurate calculations with a 2% standard error over the temperature range 78 ≤ T ≤ 2500 K [30].

$$\mu_d(T) = \frac{1.47 \times 10^{-6}\, T^{\frac{3}{2}}}{113 + T}\left[1 + 0.000153\left(\frac{T}{113} - 1\right)^2\right] \tag{30}$$

Another theoretical relationship for low density gases is given by Equation 31 which also derives from the kinetic theory of gases [19-21].

$$\mu_d(T) = a\rho_d \ell \sqrt{\frac{2k_b T}{\pi M_{rd}}} = 1.743 \times 10^{-11}\, \rho_d \ell \sqrt{T} \tag{31}$$

The constant α varies between 0.998 ≤ α ≤ 1 or can be equal to 0.667 [20]. The parameter l is the mean free path of air molecules, measured in m× $10^{-8}$, and is determined experimentally. Based on Jennings data [28], dry air with Φ = 0% and atmospheric pressure, the mean free path is

calculated by Equation 32 with a standard accuracy of 99.97% over the temperature range 288 °C ≤ T ≤ 298 °C.

$$\ell(T) = 0.0267T^2 + 14.496T \quad (32)$$

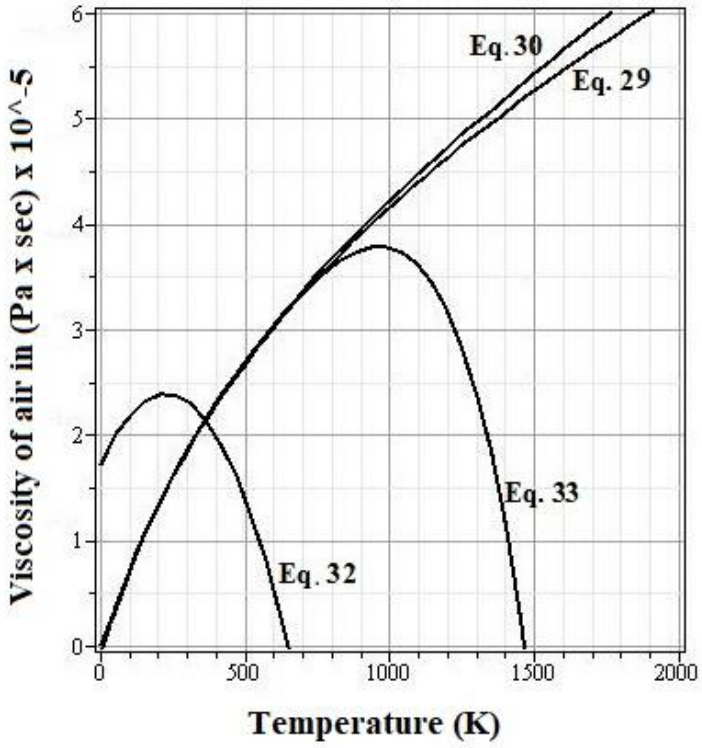



**Figure 1** The viscosity of dry air as a function of air temperature: Graphical curves based on the Equations 27, 28, 31, and 32

The viscosity of dry air can also be calculated using F.E. Jones equation, which relates temperature (in degrees Celsius), pressure (in MPa) and viscosity of air (in $N{\times}s/m^2 \times 10^{-7}$ ) [29].

$$\mu_d(T, P_d) = \mu_0(T) + \mu_0(P_d) \quad (33)$$

Where,

$$\mu_0(T) = -0.001332T^2 + 0.6054T + 170.256 \quad (33a)$$

$$\mu_0(P_d) = 0.125639{P_d}^2 + 1.17237P_d \quad (33b)$$

The polynomial equation of Irvine and Liley, p in Equation 34, is also a reliable relationship for temperature ranging from 250.15 K to 600.15 K, with air viscosity expressed in units $(N{\times}s/m^2)\times 10^{-6}$ [5- 11].

$$\mu_d(T) = M_4T^4 + M_3T^3 + M_2T^2 + M_1T + M_0 \quad (34)$$

*Determination of Water Vapor Viscosity*

Determining the viscosity of liquids through functions and relationships is more complex than for gases. Equation 35 provides a numerical relationship between the viscosity of liquids and other parameters [18,19-22]. In this equation, the ratio $N_A h_c / V_n$ equals to $2.217 \times 10^{-5}$.

$$\mu_v(T) = \frac{N_A h_c}{V_n} exp\left[\frac{3.8T_b(P)}{T}\right] = 2.217 \times 10^{-5}\, exp\left[\frac{3.8T_b(P)}{T}\right] \quad (35)$$

The boiling temperature of water $T_b$ is calculated as a function of temperature T and partial pressure using the following equation:

$$T_b(P) = \frac{7}{25}P_v + 71.6 \quad (36)$$

The viscosity of water vapor is accurately calculated using Vogel's empirical equation in Eq. (37) over a wide range of temperatures.

$$\mu_v(T) = 2.939 \times 10^{-5} \exp\left(\frac{507.88}{T - 149.3}\right) \quad (37)$$

Equation 38 is also a reliable relationship for the calculation of the viscosity of water vapor in N×s/$m^2$ × $10^{-7}$ and temperature limits 0 °C ≤ T ≤ 120 °C [5,12,13].

$$\mu_v(T) = M_5 T + M_6 \quad (38)$$

There are other formulas for calculating the viscosity of water vapor as a function of temperature, each with different standard errors and temperature limits, such as the equations of Walther and Reynolds [24] or the less practical Chapman-Enskog theory. Finally, the Reynolds number of a water vapor mixture flowing through an air duct can be expressed as a function of the density, viscosity, and velocity of each component of the mixture.

$$Re(\Phi) = f(\rho_m(\Phi), \mu_m(\Phi), u_m(\Phi)) \quad (39)$$

*The Velocity of Air – Water Vapor Mixture*

The mathematical expression of the flow velocity of an air – water vapor mixture as a function of the relative humidity of the water vapor is necessary to investigate the effect of water vapor amount on the flow velocity of the air – water vapor mixture in an air duct. The resulting equation should have the form $u_m = f(u_d(\Phi), u_v(\Phi))$. The following equations are derived from the law of conservation of mass flow rate for the fluids:

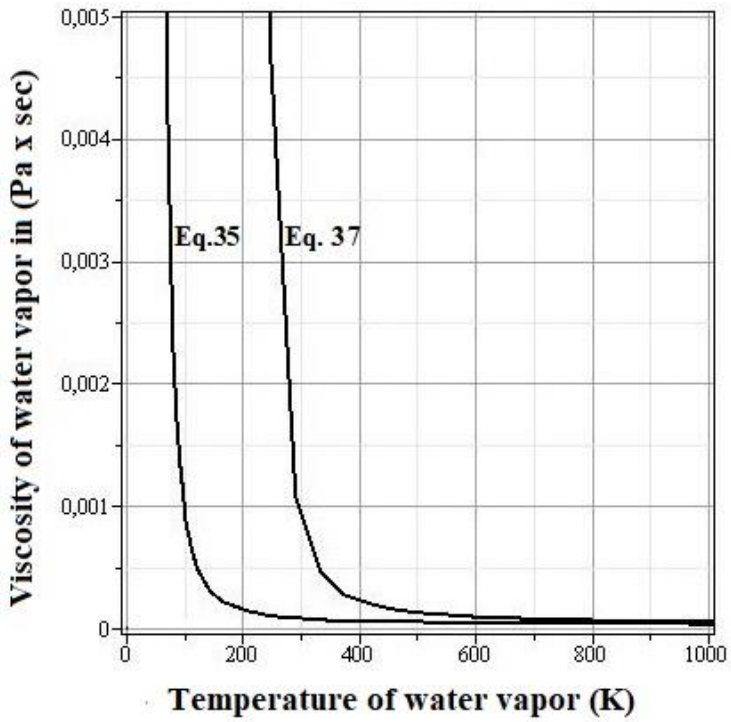



**Figure 2** The viscosity of water vapor as a function of temperature: Graphical curves based on the Equations 35 and 37

$$\dot{m_m} = \dot{m_d} + \dot{m_v} \tag{40}$$

The mass flow rate of a fluid is equal to the product of its volumetric flow rate and density at a given temperature. Additionally, according to Venturi's formula, the volumetric flow rate of a fluid is equal to the product of its flow velocity and the cross-sectional area of the air duct. Equation 40 can be expressed in the following forms:

$$\begin{aligned} \rho_m \dot{Q_m} &= \rho_d \dot{Q_d} + \rho_v \dot{Q_v} \Leftrightarrow \\ \rho_m u_m A &= \rho_d u_d A + \rho_v u_v A \Leftrightarrow \\ u_m &= \frac{1}{\rho_m}(\rho_d u_d + \rho_v u_v) \end{aligned} \tag{41}$$

The density of the air – water vapor mixture $\rho_m$ is calculated as a function of relative humidity $\Phi$ of water vapor by Equation 21. The density of dry air $\rho_d$, and water vapor $\rho_v$, may be calculated by Equations 7-9. By applying the ideal gas equation of state, Equation 41 can be expressed in the following form:

$$u_m = \frac{1}{RT\rho_m}\left(\frac{m_d P_d}{n_d} u_d + \frac{m_v P_v}{n_v} u_v\right) \tag{42}$$

By Dalton's law and Equation 13, Equation 42 takes the following final form:

$$u_m = \frac{1}{RT\rho_m}[(P_m - \Phi P_S)M_{rd}u_d + \Phi P_S M_{rv}u_v] \quad (43)$$

Where, $P_m$ is the sum of partial pressure of dry air, $P_d$ and the partial pressure of water vapor $P_v$. The partial pressure $P_v$ can be calculated using the equation of state for water vapor: $P_v = \rho_v R_v T$ where, $R_v$ is the water vapor gas constant, equal to $4.615\times 10^{-4}$ kPa $K^{-1}$ × (gr $m^{-3}$) $^{-1}$ [32].

*Application of the Mathematical Model*

The equations in Table 3 will be applied to the mathematical model used to examine the influence of the amount of water vapor on the Reynolds number of an air–water vapor mixture.

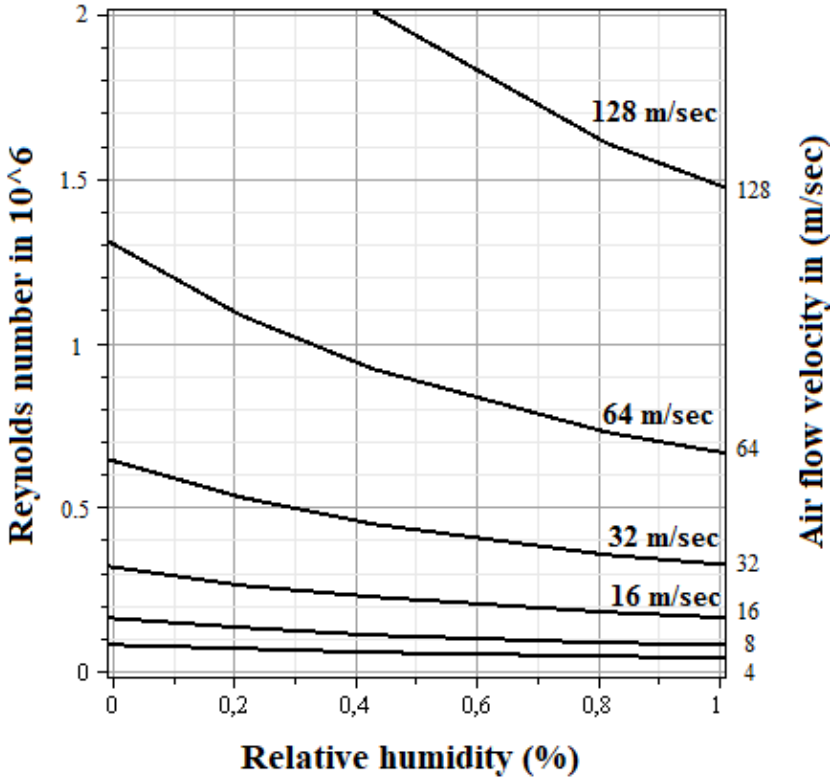



**Figure 3** The Reynolds number of an air-water vapor mixture as a function of the relative humidity of water vapor at a constant temperature T = 293.15 K and various flow velocities: u = 4, 8, 16, 32, 64 and 128 m/sec

**Table 3** Equations used to plot the curves

| Figures | Equations |
|---|---|
| Figure 3<br>Figure 5 | 1, 13, 14, 19-21, 28, 29, and 37 |
| Figure 4 | 1, 13, 16, 19-21, 28, 29, and 37 |

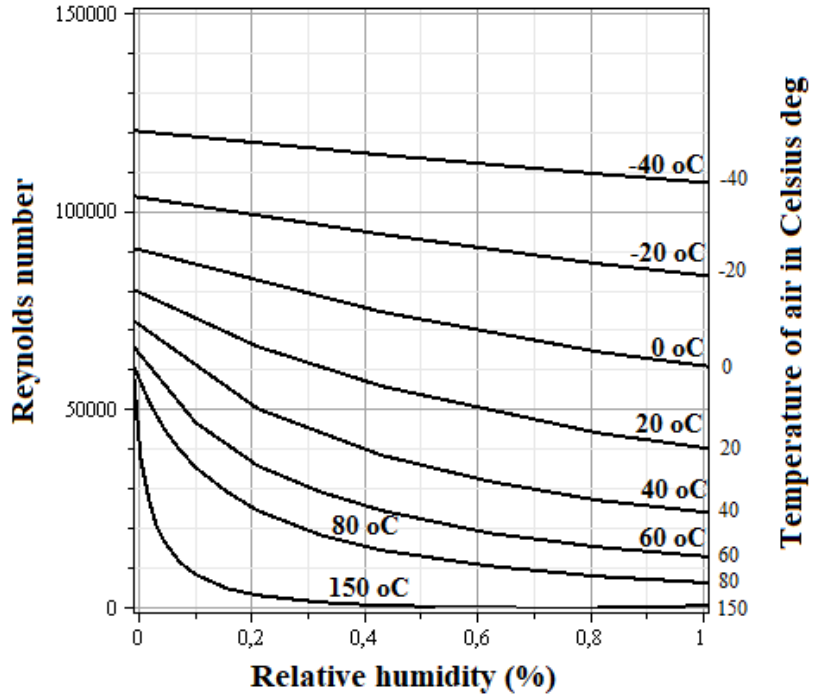



**Figure 4** The Reynolds number of an air water vapor mixture as a function of the relative humidity of water vapor at a constant flow velocity u = 4 m/sec and at various temperatures ranging from -40 to 150 °C

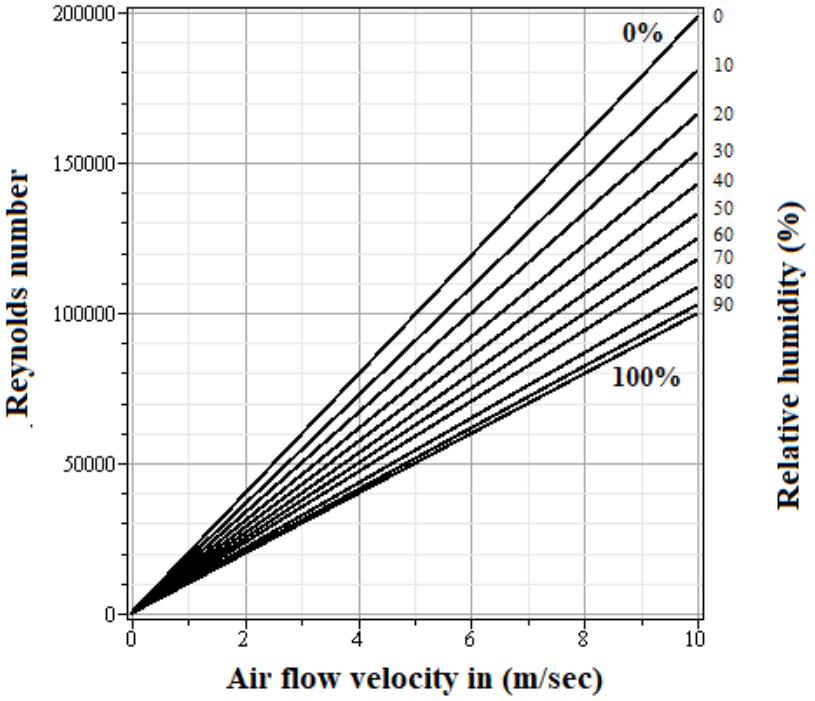



**Figure 5** The Reynolds number of an air – water vapor mixture as a function of flow velocity at a constant temperature of T = 293.15 K and at various relative humidity levels ranging from 0% to 100%

## Nomenclatures

| | |
|---|---|
| Re: | Reynolds number |
| ρ: | Density of the fluid (kg/m$^3$) |
| u: | Velocity of the fluid (m/sec) |
| μ: | Dynamic viscosity of the fluid (kg/m×sec) |
| $D_h$: | Hydraulic or equal diameter of the air duct (m) |
| Φ: | Relative humidity of mixture (%) |
| $w_s$: | Specific humidity of mixture (%) |
| x: | Mole fraction |
| n: | Amount of compound (moles) |
| m: | Mass of compound (kg) |
| $\dot{m}$ : | Mass flow rate (kg/sec) |
| M: | Molecular weight of compound (gr/mole or kg/mole) |
| P: | Pressure or partial pressure of dry air (Pa) |
| $P_S$: | Pressure of saturated water vapor of the mixture (Pa, hPa, or kPa depending on the formula) |
| $P_Q$: | Supply pressure of the pump (Pa) |
| T: | Temperature of the fluid, air or liquid (°C or K) |
| $T_\theta$: | Temperature of water vapor (°C) |
| $T_b$: | Boiling temperature of water (°C) |
| $c_w$: | Dimensionless Wilke's coefficient. |
| ℓ: | Mean free path of air molecules (kg×10$^{-8}$) |
| $\dot{Q}$ : | Volumetric flow rate (m$^3$/sec) |
| A: | Internal section of air duct (m) |

### *Indicators*

The following indicators are related to the physical quantities of density, viscosity, velocity, pressure, mass of compound, mass and volumetric flow rate, molecular weight, and the amount of the compound.

| | |
|---|---|
| T: | Mixture from n individual components |
| m: | Air – water vapor mixture |
| d: | Dry air |
| v: | Water vapor |
| 0: | Viscosity of dry air in F. E. Jones equation |
| rd: | Molecular weight of dry air |
| rv: | Molecular weight of water vapor |

*Constants*

| | |
|---|---|
| R: | Gas constant (8.314 J/mole×K) |
| $R_v$: | Water vapor gas constant ($4.615\times10^{-4}$ kPa $K^{-1}$ (gr $m^{-3}$) $^{-1}$) |
| α: | Equal with 0.667 or $0.998 \leq \alpha \leq 1$ |
| $a_0$: | 1.131439334 |
| $a_1$: | $-3.750393331\times10^{-2}$ |
| $a_2$: | $5.591559189\times10^{-3}$ |
| $a_3$: | $-6.220459433\times10^{-5}$ |
| $a_4$: | $1.10581611\times10^{-6}$ |
| π: | 3.14158 |
| $k_b$: | Boltzmann constant ($1.3806\times10^{-23}$ J/K) |
| $N_A h_c/V_n$: | $2.217 \times 10^{-5}$ |
| $M_0$: | $1.685731754 \times 10^{-5}$ |
| $M_1$: | $9.151853945 \times 10^{-8}$ |
| $M_2$: | $-2.16276222 \times 10^{-9}$ |
| $M_3$: | $3.413922553 \times 10^{-11}$ |
| $M_4$: | $-2.644372665 \times 10^{-13}$ |
| $M_5$: | 0.4 |
| $M_6$: | 80.58131868 |
| $P_{atm}$: | Atmospheric pressure (101293 Pa). |

*Abbreviations*

| | |
|---|---|
| HEPA: | High efficiency particulate air filter. |
| UVGI: | Ultraviolet germicidal irradiation lamp. |
| CFD: | Computational fluid dynamics. |

## Discussion

The application of the resulting mathematical model and the analysis of the variation curves demonstrate that a mathematical relationship exists between the Reynolds number of the air–water vapor mixture and the relative humidity of the water vapor within the mixture. Specifically, changes in the amount of water vapor alone

affect the velocity, density, and viscosity of the flowing mixture, assuming air is considered as a two-phase air–water vapor mixture. As a result, both flow velocities, air density, viscosity, and relative humidity can influence the Reynolds number of the mixture as it flows through a horizontal air duct.

The flow of a two-phase mixture through an air duct is a process that can occur without the need for additional interventions to alter the Reynolds number of the flowing air. Such interventions might include adjusting the air velocity or temperature of the mixture using an air valve, heating element, or cooling device. A change in relative humidity alone is sufficient to affect the Reynolds number of the flow. Figures 3, 4, and 5 show that it is possible to achieve a targeted variation ΔRe in the Reynolds number of the air–water vapor mixture with good accuracy, as a function of changes in the relative humidity ΔΦ of the water vapor. This means that an operator of an experimental apparatus or industrial system can control the Reynolds number of the flow by adjusting the relative humidity of the water vapor. The mathematical model presented in this study can be applied in various situations that require controlled conditions for water vapor content and Reynolds number (see Tables 1 and 2).

Figure 3 shows that an increase in the relative humidity of water vapor leads to a decrease in the Reynolds number of the air–water vapor mixture. This decrease is especially significant at higher flow velocities ($u > 16$ m/s) compared to lower flow velocities ($u < 16$ m/s). At flow velocities of 4 and 8 m/s, the Reynolds number decreases almost linearly. Figures 4 and 5 also demonstrate that as the relative humidity of water vapor increases, the Reynolds number of the air–water vapor mixture decreases. This conclusion is supported by the studies of Iskra and Simonson [49], and likely by the findings of Westhoff et al. [52]. However, there are differing conclusions among the studies. Four studies [45-48] report that an increase in the amount of water vapor in the air–water vapor mixture leads to an increase in the Reynolds number. In contrast, two studies by the same group of researchers [49,50] conclude that the Reynolds number decreases as the water vapor content increases.

## Conclusion

An analysis of the variation curves of the Reynolds number concerning the effect of water vapor content on airflow leads to the following conclusions: As the relative humidity of water vapor increases, the Reynolds number of the air–water vapor mixture decreases during flow through an air duct, across various air temperatures and flow velocities. In some cases, the Reynolds number is not significantly affected by changes in flow velocity. Additionally, increasing the relative humidity from 0% to 100% results in a reduction of the Reynolds number by up to 50%, regardless of fluctuations in flow velocity, when the mixture is at a constant temperature of 293.15 K. Regarding the effect of temperature: over a range of flow velocities, the relative humidity exhibits a linear relationship with the Reynolds number when the temperature ranges from –40 °C to 0 °C, and a nonlinear relationship at temperatures above 20 °C. Furthermore, when the air–water vapor mixture is at temperatures higher than 150 °C and the relative humidity ranges from 0% to 20%, the Reynolds number decreases rapidly. Additionally, the Reynolds number of the air–water vapor mixture varies linearly with flow velocity, regardless of the amount of water vapor present in the mixture.

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## Conflicts of Interest

The authors declared no conflict of interest in this study.

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